

Head & Hand

AI Career & Interview Coach

HCAI Term Project — Milestone 2

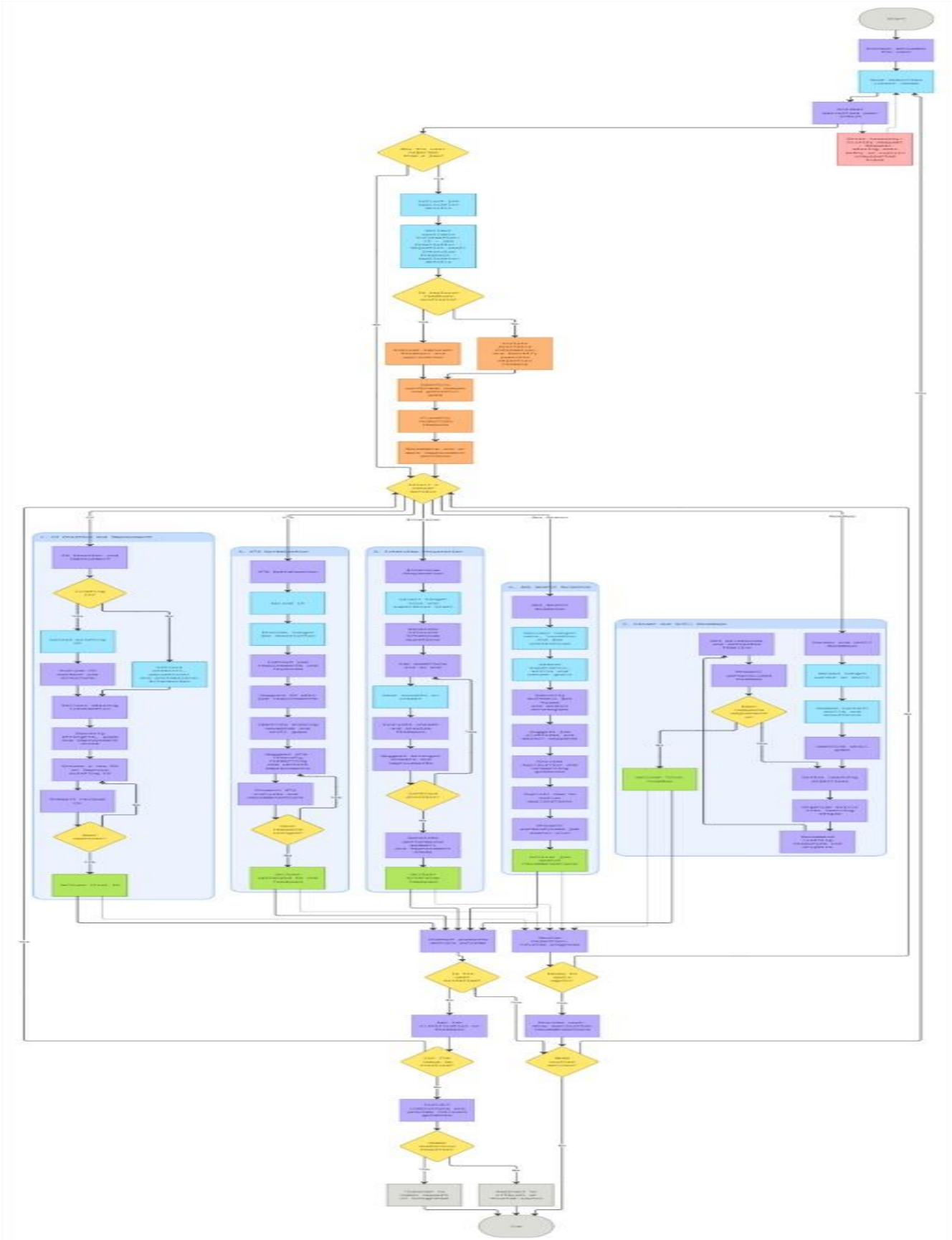
HCAI Principle: User Control and Agency

TEAM MEMBER:

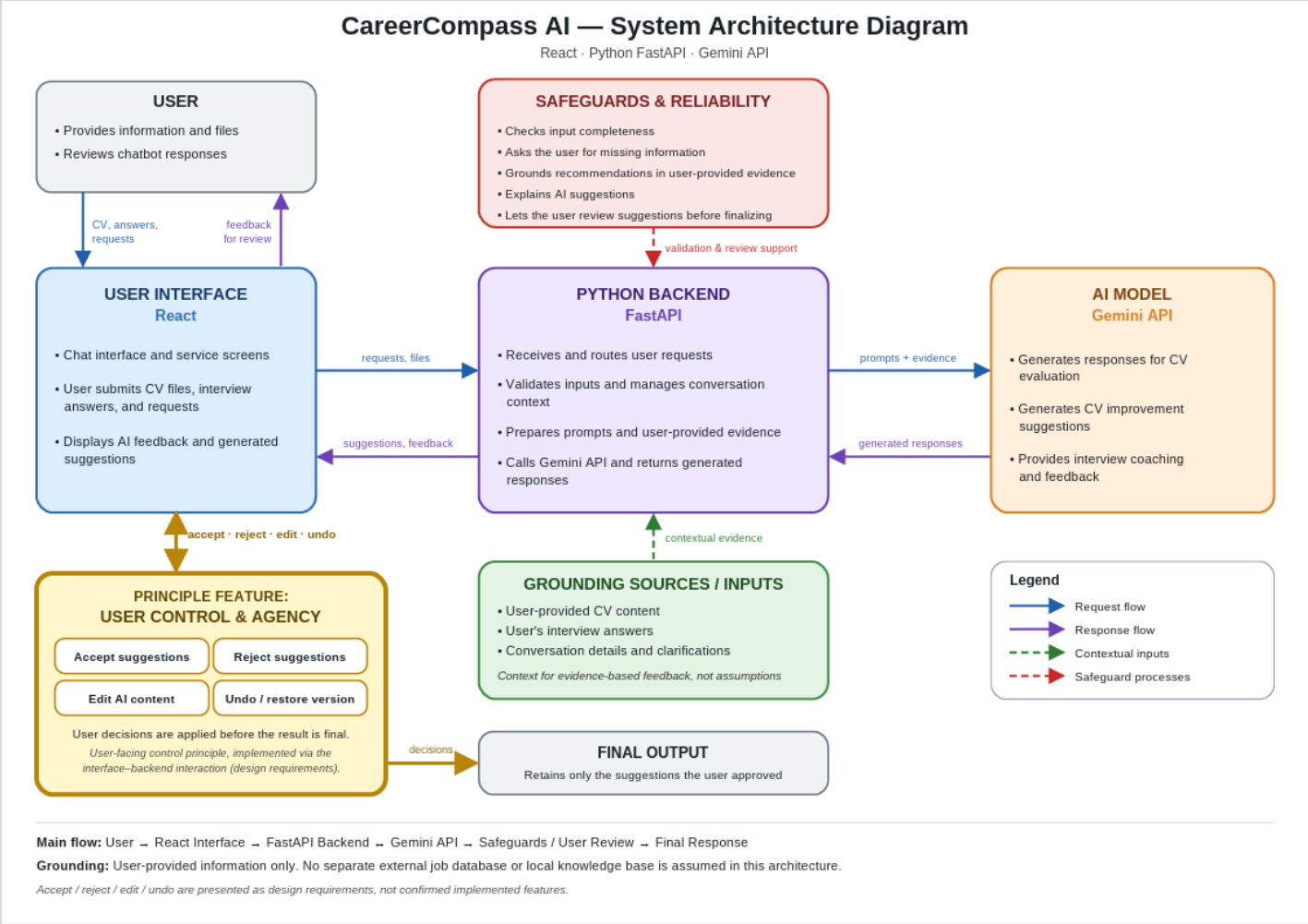
- Nora Walid Ibrahim
- Maisoun Ebrahim Ahmed
- Noor Adal Mohammed
- Fatma Saber Elgendy

Task 1: Conversation Flow Diagram

[Link of workflow]



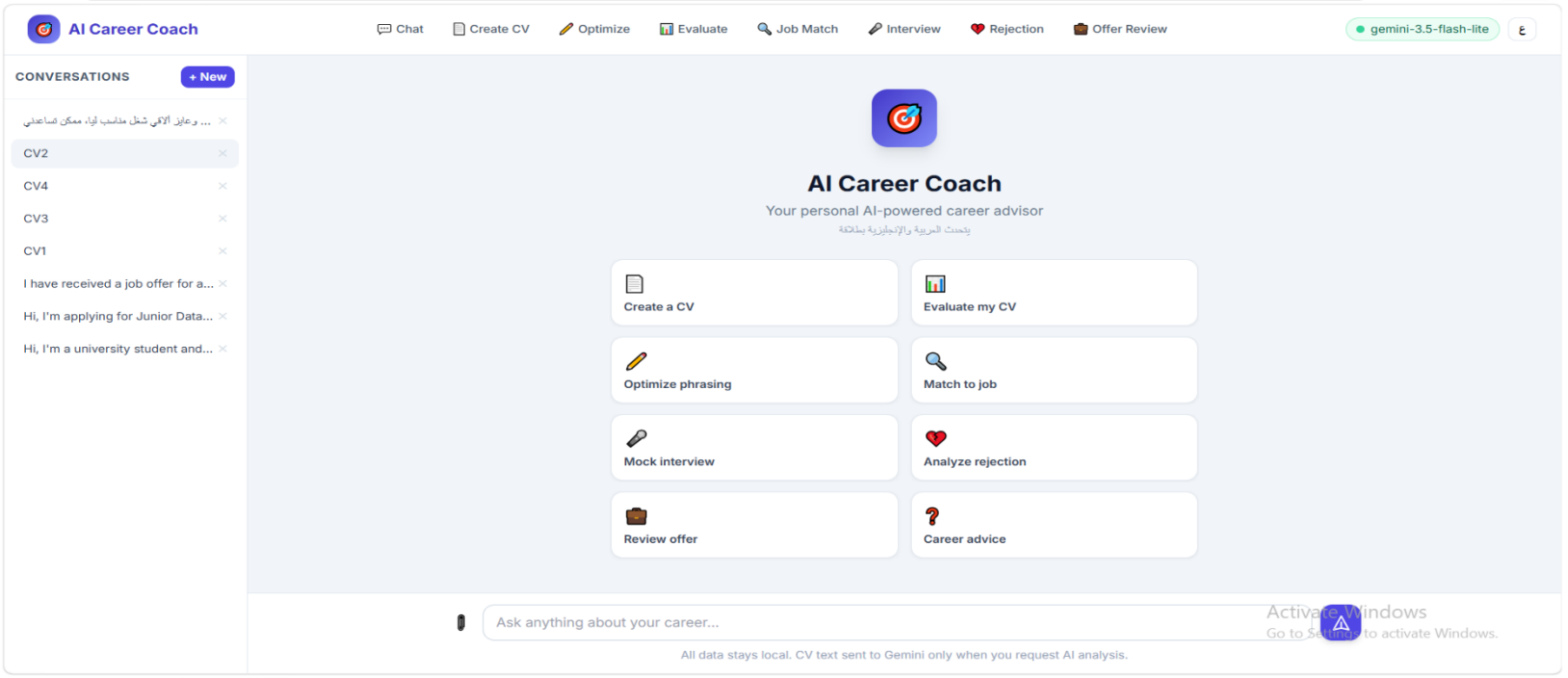
Task 2: System Architecture Diagram

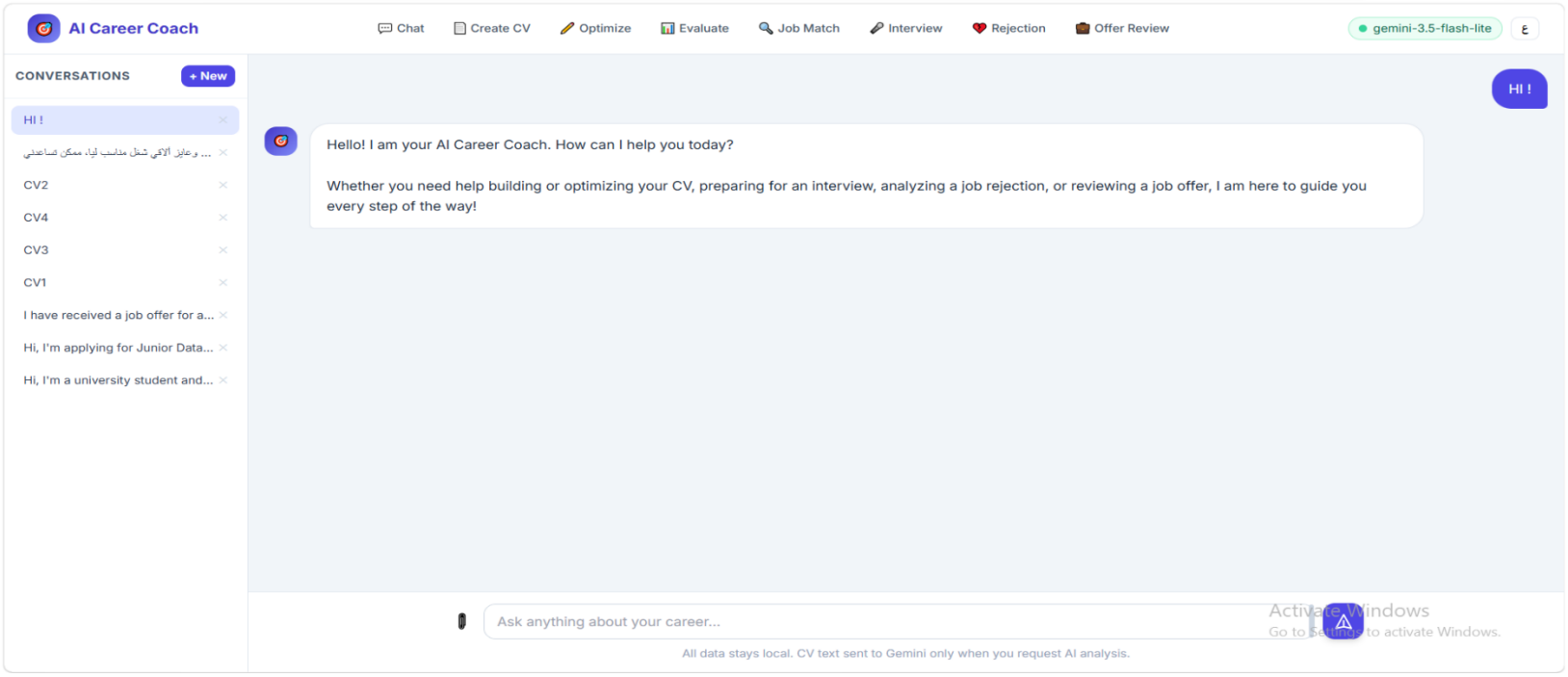


Task 3: Shneiderman's 8 Golden Rules Checklist

#	Golden Rule	Application in CareerCompass AI
1	**Strive for Consistency**	Use consistent colors, typography, button styles, icons, and interaction patterns across the chatbot interface and all career service screens, including CV evaluation, CV improvement, and interview coaching.
2	**Enable Shortcuts**	Provide quick replies, suggested prompts, and reusable actions for common requests, such as uploading a CV, requesting CV feedback, or starting interview practice, reducing repetitive user input.
3	**Offer Informative Feedback**	Display clear status messages during operations, such as CV uploading, document processing, AI evaluation, and response generation. Show a completion message when each task finishes.
4	**Design Dialogs to Yield Closure**	After completing a task, present a clear summary of the result, such as CV evaluation findings, suggested improvements, or interview performance feedback, and indicate the next available action.
5	**Prevent Errors and Handle Them Simply**	Validate uploaded file types and required inputs before processing. If information is missing, a file is unsupported, or an AI request fails, explain the problem clearly and guide the user toward a correction or retry.
6	**Permit Easy Reversal of Actions**	Allow users to undo previous edits or restore an earlier version of AI-generated CV content, where version history is supported. Users should also be able to revise their choices before finalizing suggestions.
7	**Support Internal Locus of Control**	Keep users in control of AI-generated recommendations by allowing them to accept, reject, or edit suggestions. The chatbot should explain its recommendations and ask for clarification instead of making unsupported assumptions or final decisions on the user's behalf.
8	**Reduce Short-Term Memory Load**	Display relevant instructions, required information, suggested actions, and previous conversation context within the interface. Use guided steps and visible prompts so users do not need to memorize requirements while completing career-related tasks.

Task 4: First Model Call from Your Own Backend





[Link of Recorded chat]

Task 5: Prompt Log – Appendix A

Prompt of Conversation Flow Diagram

Create a complete, professional, editable Functional Workflow Flowchart for CareerCompass AI based on the provided workflow.

Requirements:

- Use a top-down (TD) layout.
- Preserve all workflow steps, decision points, Yes/No branches, connections, and feedback loops.
- Do not remove or invent any functionality.
- Use rectangles for processes, diamonds for decisions, and rounded rectangles for Start/End.
- Organize the five services into clearly separated sections.
- Ensure readable labels, clear arrows, and minimal overlapping.
- Use a professional academic design with consistent colors.

Main Workflow Sections:

1. Initial Engagement: Start, chatbot welcome, user career needs, and intent identification.
2. Job Rejection Analysis: Collect application information, analyze employer feedback or available data, identify rejection reasons, and recommend improvement services.
3. Service Selection: Five main services:
    - CV Creation and Improvement.
    - ATS Optimization.
    - Interview Preparation.
    - Job Search Guidance.
    - Career and Skill Roadmaps.
4. Service Workflows: Include all steps within each service, including information collection, analysis, recommendations, user review, approval, and final delivery.
5. User Satisfaction: Present service outcomes, check user satisfaction, handle feedback, and allow selecting another service or ending the session.
6. Error Recovery: Clarify requests, collect missing information, retry, or explain limitations.
7. Rejection Improvement Loop: Review progress, check readiness to apply again, and provide next-step recommendations.

Color Coding:

- Light Blue: User actions.
- Light Purple: Chatbot/AI actions.
- Light Yellow: Decisions.
- Light Green: Successful outcomes.
- Light Orange: Rejection analysis.
- Light Red: Error handling.
- Light Gray: Start/End.

Important: Maintain the complete logical workflow, including all five service subgraphs, Yes/No branches, approval loops, interview practice loops, error recovery, and rejection-related improvement connections.

Use a large landscape canvas if needed. The final result must be an actual editable flowchart, not an image or textual explanation.

Prompt of System Architecture Diagram

Subtitle: React · Python FastAPI · Gemini API

Create a professional, clean, editable **System Architecture Diagram** using a landscape layout, white background, rounded boxes, readable text, and color-coded arrows.

1. Main Components

- **USER (Gray):** Provides information and files; reviews chatbot responses.
- **USER INTERFACE – React (Light Blue):** Chat interface, service screens, CV uploads, interview answers, requests, and AI feedback.
- **PYTHON BACKEND – FastAPI (Light Purple):** Routes requests, validates inputs, manages conversation context, prepares prompts, calls Gemini API, and returns responses.
- **AI MODEL – Gemini API (Light Orange):** Generates CV evaluations, CV improvement suggestions, and interview coaching and feedback.
- **SAFEGUARDS & RELIABILITY (Light Red):** Checks missing information, grounds recommendations in user evidence, explains suggestions, and supports user review.
- **GROUNDING SOURCES / INPUTS (Light Green):** User-provided CV content, interview answers, conversation details, and clarifications.
  Note: Context for evidence-based feedback, not assumptions.

2. User Control & Agency (Pale Yellow)

Create a prominent box containing four controls in a 2×2 grid:

- Accept suggestions
- Reject suggestions
- Edit AI content
- Undo / restore version

Text: "User decisions are applied before the result is final."

Note: User-facing control principle, implemented via interface-backend interaction (design requirements).

3. Final Output (Gray)

FINAL OUTPUT

"Retains only the suggestions the user approved"

4. Connections and Arrows

- User → Interface: Blue – CV, answers, requests.
- Interface → User: Purple – feedback for review.
- Interface ↔ Backend: Blue requests/files; Purple suggestions/feedback.
- Backend ↔ Gemini: Blue prompts + evidence; Purple generated responses.
- Safeguards → Backend: Red dashed – validation & review support.
- Grounding Sources → Backend: Green dashed – contextual evidence.
- Interface ↔ User Control: Gold double-headed arrow – accept, reject, edit, undo.
- User Control → Final Output: Gold – decisions.

5. Legend

- Blue solid: Request flow.
- Purple solid: Response flow.
- Green dashed: Contextual inputs.
- Red dashed: Safeguard processes.

6. Footer

Main flow: User → React Interface → FastAPI Backend → Gemini API → Safeguards / User Review → Final Response.

Grounding: User-provided information only. No external job database or local knowledge base is assumed.

Accept / reject / edit / undo are design requirements, not confirmed implemented features.

Design Instructions

Maintain all components, labels, arrows, and logical connections. Use balanced spacing, clear typography, and a professional academic style. Do not add databases, external services, or unrequested features.